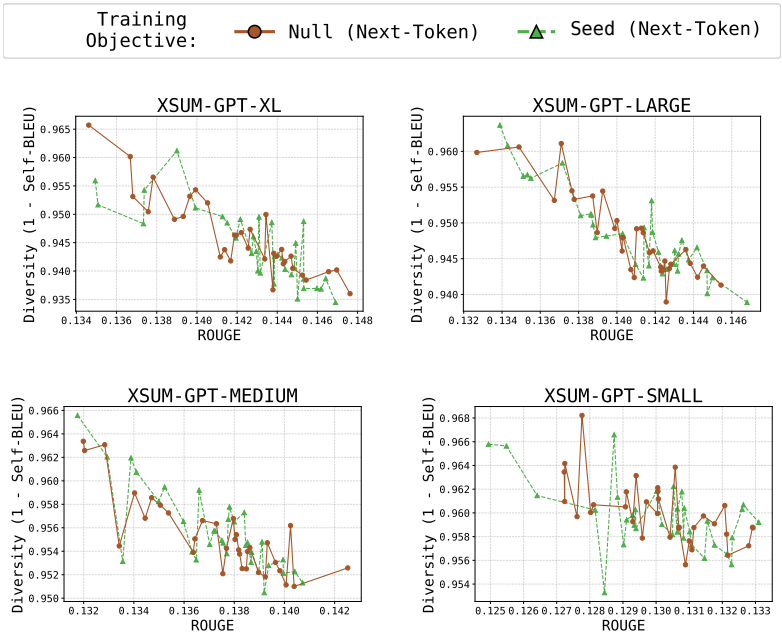

Figure 35. **Seed-conditioning has no effect on diversity for `GPT` models on `XSUM` summarization with next-token prediction.**

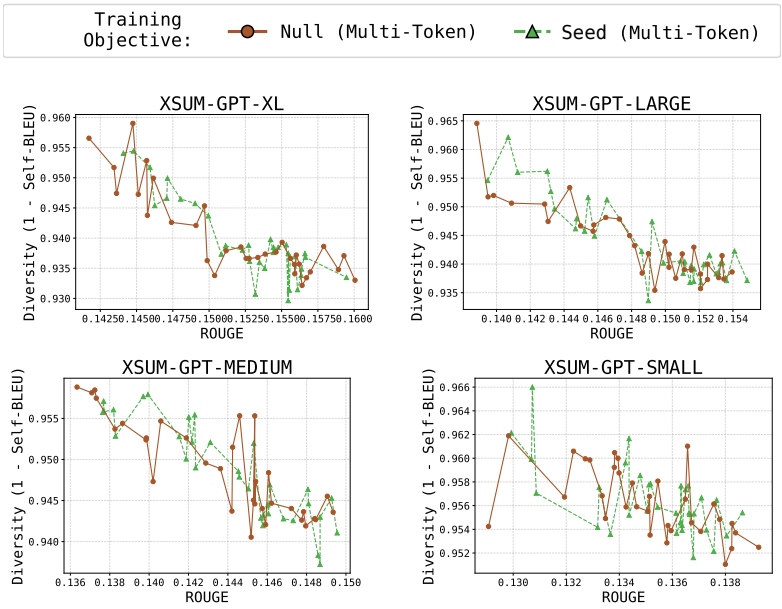

Figure 36. **Seed-conditioning has no effect on diversity for `GPT` models on `XSUM` summarization with multi-token prediction.**

J. More related works

Empirical studies of creativity in LLMs. There is a long line of recent works that measure novelty and creativity of LLMs and LLM-assisted users. (Chakrabarty et al., 2024; Lu et al., 2024b) quantitatively evaluate and report that models vastly underperform under expert human evaluation against human writers. Zhang et al. (2024a) argue that finetuning methods such as RLHF and DPO, are limited when applied to creative humor-generation tasks. Likewise models like GPT4 and Claude currently underperform top human contestants in generating humorous captions. In poetry, Walsh et al. argue that there are certain characterstic styles that ChatGPT restricts itself to. Even assisted-writing can reduce diversity

(Padmakumar & He, 2024) or produce bland writing (Mirowski et al., 2024). On the positive side, Si et al. (2024) report that LLMs surprisingly generate novel research ideas, although these are less feasible. Anderson et al. (2024) find that users tend to produce more divergent ideas when assisted by ChatGPT (although at a group level, ideas tend to homogenize). Another line of works (Wang et al., 2024a; Talmor et al., 2020; Zhong et al., 2024) has proposed algorithmic improvements involve creative leaps-of-thought for real-world tasks.

Other studies have proposed benchmarks for evaluating creativity. AidanBench (McLaughlin et al., 2024) and NoveltyBench (Zhang et al., 2025) evaluate LMs on their ability to produce diverse and coherent responses by penalizing repetition across generations. However, they do not measure originality relative to training data, leaving open whether outputs are genuinely novel or simply unseen paraphrases/recombinations. Zhao et al. (2024) evaluate LM creativity using the Torrance Tests of Creative Thinking, a standard in human psychometrics. Another line of work such as Alchemy (Wang et al., 2021), IVRE (Xu et al., 2022), and DiscoveryWorld (Jansen et al., 2024) present simulations with hidden facts and rules, requiring LMs to explore, hypothesize, and test through interaction. While these simulations focus on pretrained models rather than examining how training shapes creative capabilities, they serve as valuable and realistic benchmarks for assessing the role of creativity in scientific discovery.

Finally, we refer the reader to Franceschelli & Musolesi (2023) for a rigorous treatment of philosophical questions surrounding creativity in LLMs. We also a refer to Wang et al. (2024b) for a theoretical treatment of how to formalize subjectivity in creativity.

The next-token prediction debate. In support of next-token prediction, there are arguments (Shannon, 1948; 1951; Alabdulmohsin et al., 2024) that claim that language is captured by NTP with models even superceding humans (Shlegeris et al., 2022) at NTP. There are also theoretical results emphasizing the expressivity (Merrill & Sabharwal, 2024; Feng et al., 2023) and learnability (Malach, 2023; Wies et al., 2023) of autoregressive Transformers as long as there is a sufficiently long chain of thought.

Multi-token training. While multi-token methods employ diverse strategies, a common feature is their reliance on objectives that capture dependencies across entire sequences. Representative examples include teacherless training (Bachmann & Nagarajan, 2024; Monea et al., 2023; Tschannen et al., 2023) and independent output heads or modules (Gloeckle et al., 2024; DeepSeek-AI et al., 2024) or inserting a lookahead attention (Du et al., 2023). Another line of research is discrete diffusion models (Hoogeboom et al., 2021; Austin et al., 2021; Gong et al., 2023; Lou et al., 2023), which avoid strict left-to-right factorization by iteratively refining an entire sequence at multiple positions. There are other models as well, such as energy-based models (Dawid & LeCun, 2023) and non-autoregressive models or (Gu et al., 2018). Frydenlund (2024) report gains from non-autoregressive models on the path-star task (Bachmann & Nagarajan, 2024), while Hu et al. (2025) propose a novel method call the Belief State Transformer that shows gains on these tasks. While we only report the effect of teacherless training on our tasks due to its simplicity, we speculate that these methods would show similar (or better) results as teacherless training in our settings.

Transformers and graph algorithmic tasks. Graph tasks have been used to understand various limitations of Transformers in orthogonal settings. Bachmann & Nagarajan (2024); Saparov et al. (2024) report that Transformers are limited in terms of learning to search tasks on graphs, while Sanford et al. (2024) provide positive expressivity results for a range of algorithmic tasks that process an graph. These works differ from our study of combinational creativity since their graphs are provided in-context and the tasks have a unique answer. Other works (Schnitzler et al.; Yang et al., 2024a;b) study multi-hop question answering on a knowledge graph; however, this does not require planning.

Diversity of generative models. One line of work relevant to us in the history of generative models is RNN-based VAE for text data (Bowman et al., 2016). The motivation, like in our work, was to learn high-level semantic features rather than next-token features with the hope of producing more novel sentences. However, this suffered from posterior collapse, where the model ignores the latent variable altogether inspiring various solutions (Yang et al., 2017; Goyal et al., 2017). Our results on seed-conditioning are also reminiscent of a line of work on exploration in reinforcement learning (RL), where it has been shown that adding noise to the policy model parameters enables more efficient exploration than directly adding noises to the output space (Plappert et al., 2017; Fortunato et al., 2017).

Learning-theoretic studies of diversity in LLMs. Various theoretical works provide rigorous arguments for how preventing hallucination and maximizing the model's coverage are at odds with each other in abstract settings (Kalai &

Vempala, 2024; Kalavasis et al., 2024; Kleinberg & Mullainathan, 2024). We clarify that this tension does not apply in our concrete settings. In those abstract settings, the strings in the support can be arbitrary and adversarially chosen whereas, our strings are generated by a simple rule (which can be learned).

Another theoretical question underlying generative models is that the optimum of their objectives are attained at perfect memorization; yet they tend to produce novel examples e.g., this question has been posed for GANs in Nagarajan et al. (2018) and for diffusion in Nakkiran et al. (2024) (see "remarks on generalization") or Kamb & Ganguli (2024). Okawa et al. (2023); Kamb & Ganguli (2024) study the creativity ability of image diffusion models in combining various concepts to create an image. Unlike our setting, these do not require planning.

Injecting input noise. We note that the concept of injecting input noise into a Transformer has been explored before, but for other functions (such as quality, robustness (Hua et al., 2022; Jain et al., 2024) or efficiency (Wang et al., 2024c).), and in other forms (e.g., inducing Gaussian noise).